

NPTEL Course on
"Computer System Optimization for ML"

Lecture #54

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Agenda

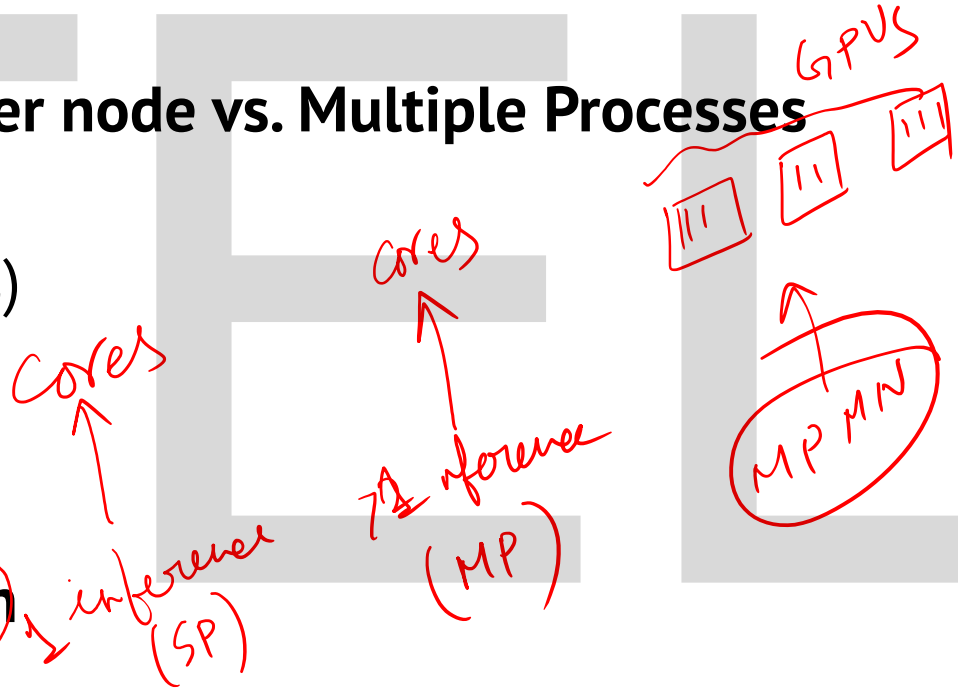
- **Performance Characterization of Deep Learning Workloads (contd.)**

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1. Inference
2. Inf + Train (GPU)

Performance Characterization of DNN Training [1]

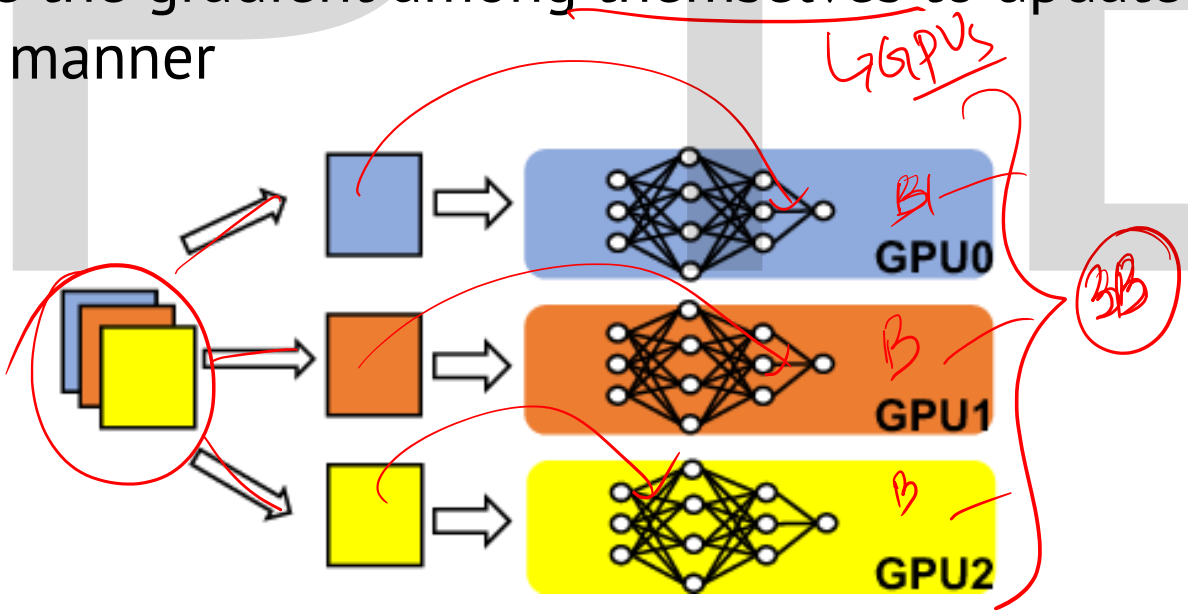
- **Impact of #cores of a CPU**
- **Impact of Batch Size (BS)**
- **Single Process (SP) vs. Multiple Process (MP) per node vs. Multiple Processes and Multiple Nodes (MPMN)**
  - On Clusters with varying size (#nodes, #cores)
- **Relation between clock speed and core count**
- **CPU and GPU performance**
- **Performance trends for TensorFlow and PyTorch**



[1] Jain, Arpan, et al. "Performance characterization of dnn training using tensorflow and pytorch on modern clusters." *2019 IEEE International Conference on Cluster Computing (CLUSTER)*. IEEE, 2019.

Recap: Data Parallelism in DNN Training

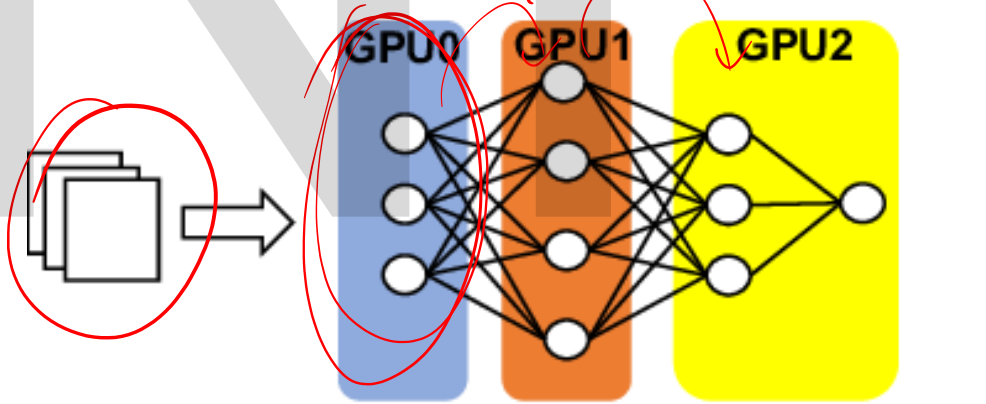
- **Data Parallelism**
  - Training data is distributed across processes
  - DL model is replicated and executed on each process
  - Batches across processes together form a larger batch
  - Communicate the gradient among themselves to update the weights in a synchronous manner



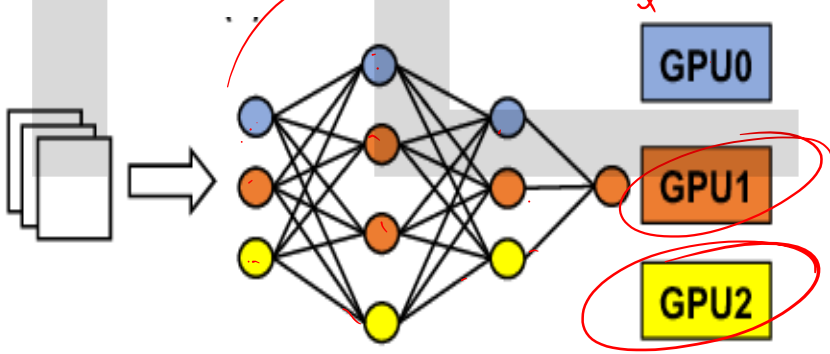
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Recap: Model Parallelism in DNN Training

- **Model Parallelism**
  - DL model split across processes (at layer granularity)
  - Send and Receive communication operations for passing the feature maps between layers (in both forward and backward passes)



(b) Model Parallelism—layer-wise

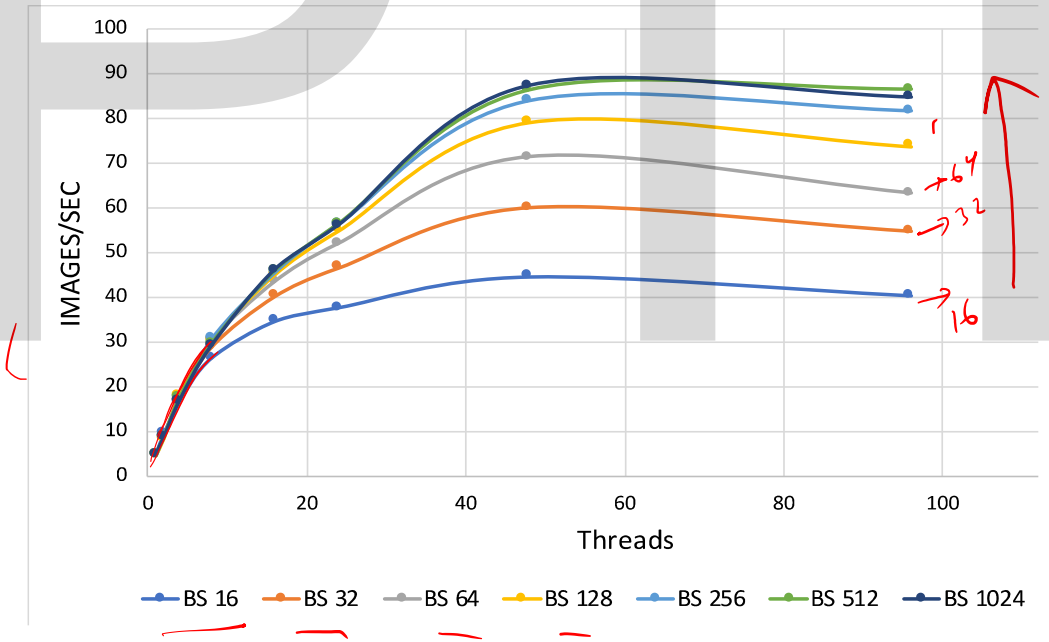


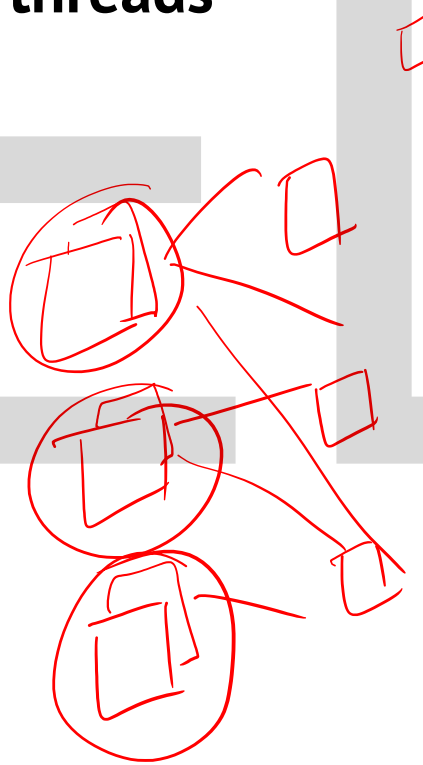
(b) Model Parallelism—tensor slicing

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Single Process (SP) Performance

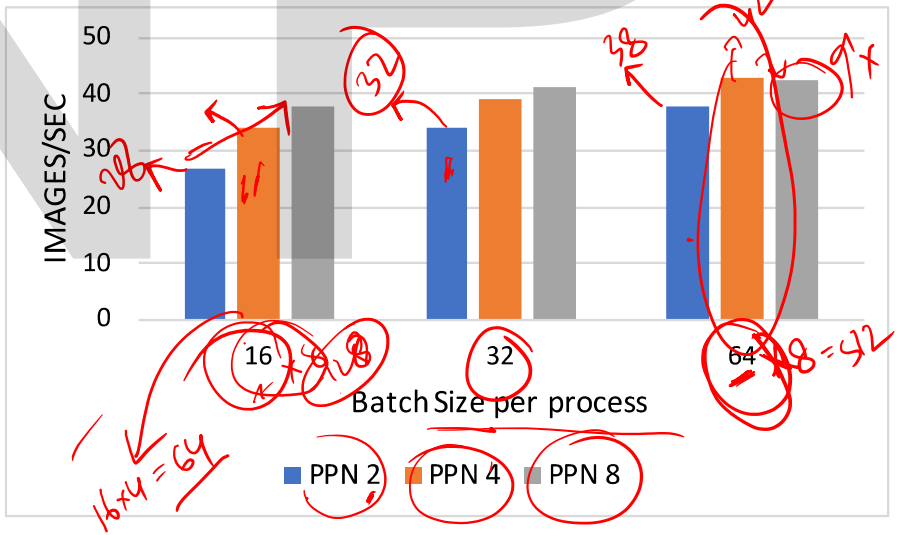
- Parallelism exploited is intra-op in the single node
- Perf. Improves with no. of threads (upto 48) on a Skylake 48 core system!
- Larger BS can achieve higher improvement with increase in threads

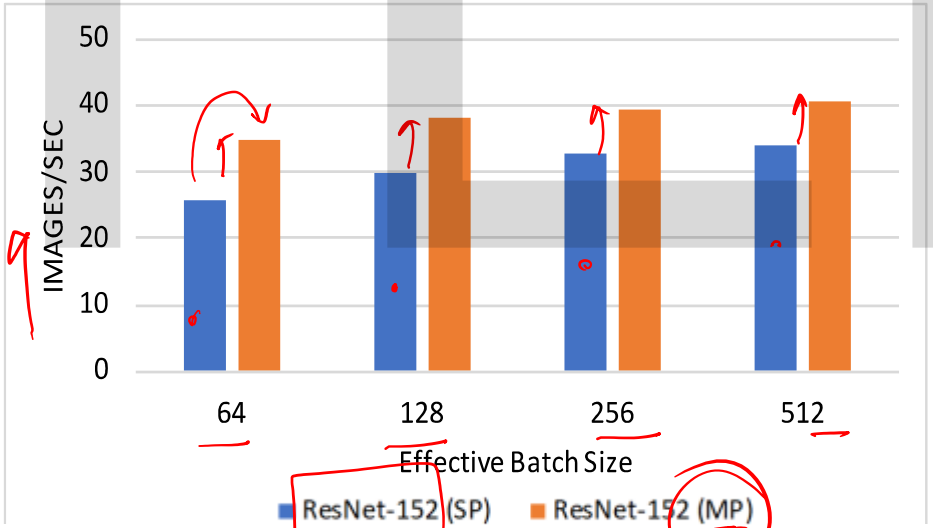




Multiple Process (MP) Performance

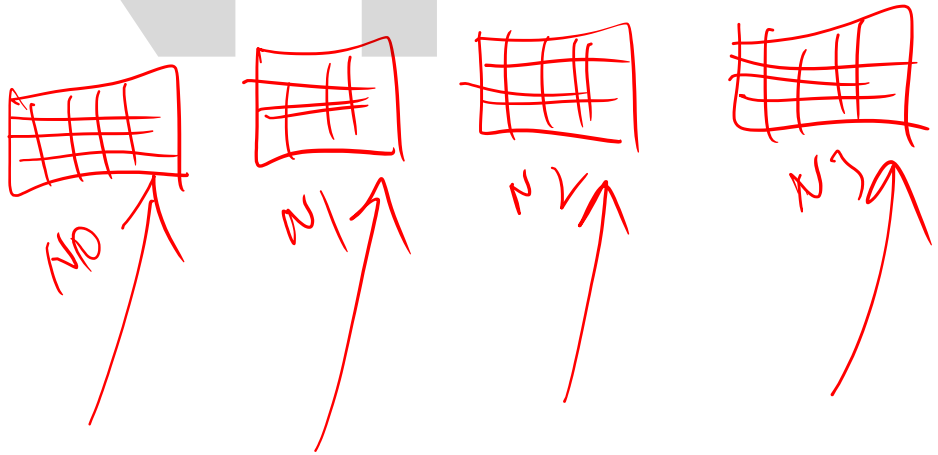
- Distributed training within a node (ResNet-152)
- Memory per process limits process per node (PPN)
- 4 PPN gives good performance up to BS=64
- MP gives better performance than SP in a node

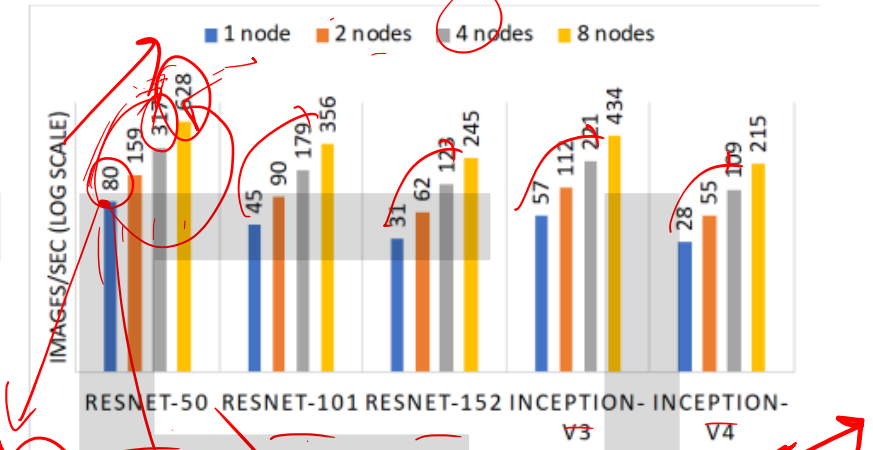


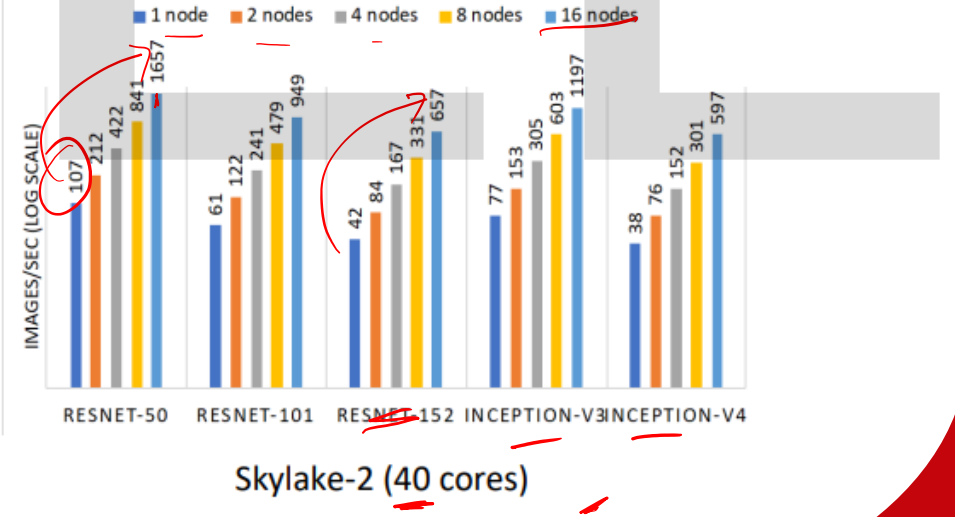


MultiNode MultiProcess (MPMN)

- PPN fixed based (best) MP performace
- No. of threads PPN tuned to give best training performance
- Scalable performance on MPMN
- Similar trend for all applications
- Similar trends across Tensorflow and Pytorch

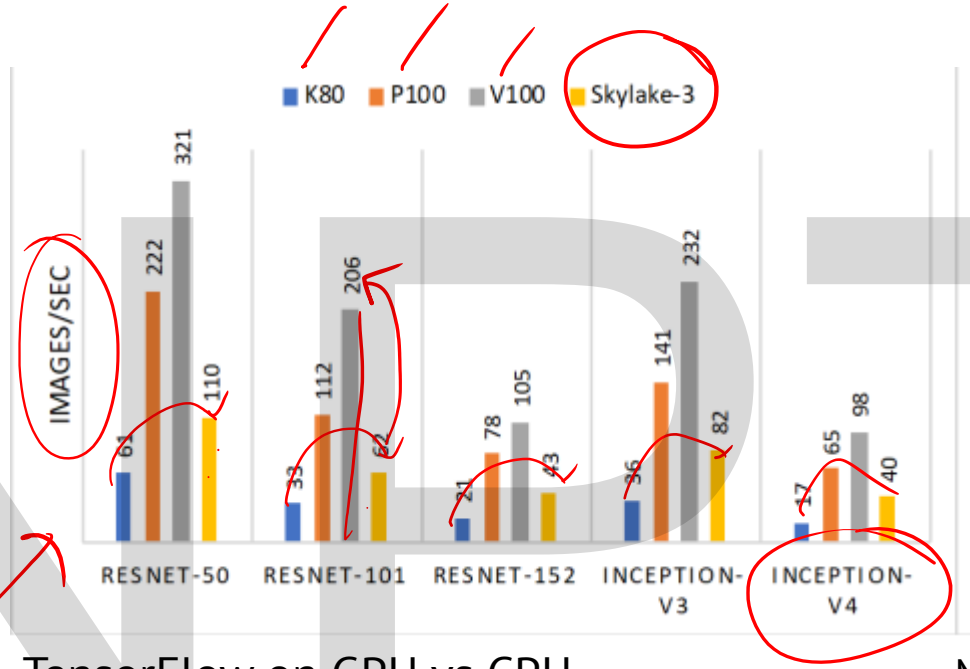


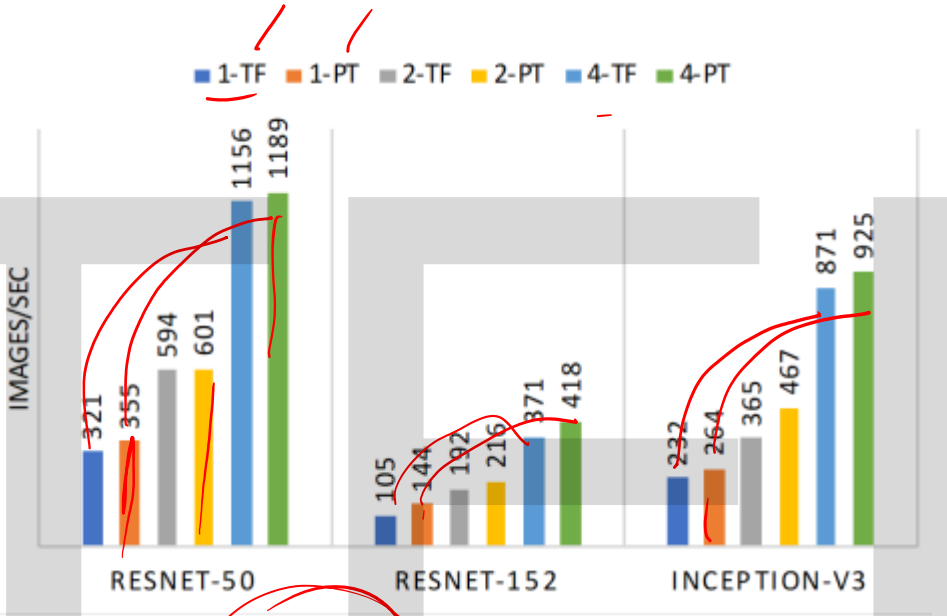




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TensorFlow and PyTorch: CPU vs. GPU





TensorFlow on GPU vs CPU

- Inception-v4: Skylake-3 is up to 2.35× faster than K80
- ResNet-101: V100 is up to 3.32× faster than Skylake-3

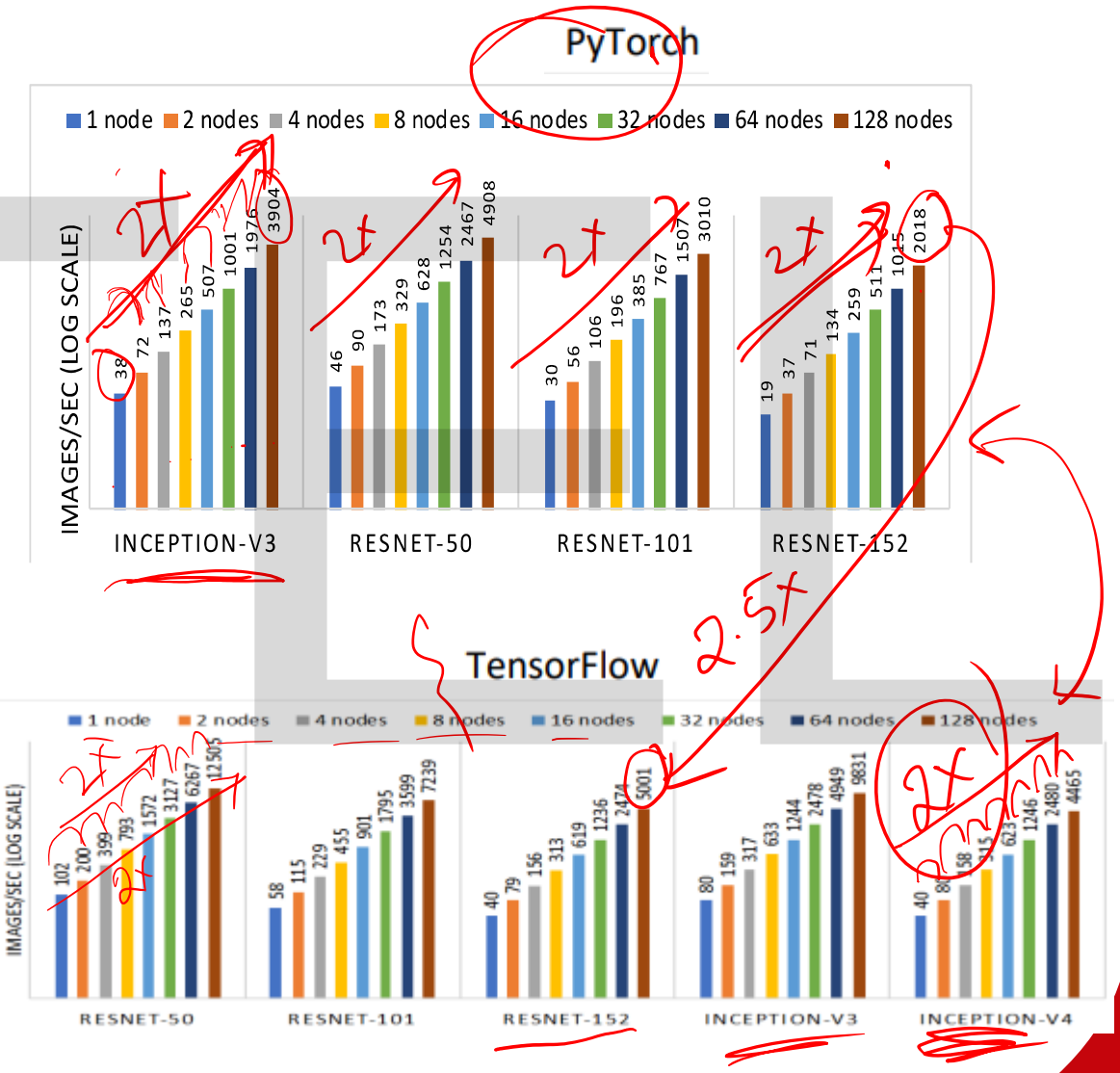
Multi-node: TF vs PT

- ResNet-50: PT is slightly better than TF
- ResNet-152: PT is up to 12% better than TF

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PyTorch vs TensorFlow

- **Similar trend across PyTorch and TensorFlow**
- **TensorFlow is up to 2.5× faster than PyTorch for 128 nodes**
- **PyTorch scales well but overall lower performance than TensorFlow**
  - Back in 2019





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)